
What does a language model detect when it detects an injected thought?

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Abstract

Language models are reported to detect when a concept vector has been injected into their residual stream. Two critiques argue that the report reflects generic sensitivity to disturbance rather than access to content, one inferring this from the model’s confabulations and one from its confusion between input and activation manipulations. The defended mechanistic account controls only against the absence of injection. We run the manipulation both critiques point toward but did not perform: on the defended model, layer, and prompt (Gemma3-27B-it, layer 37), we compare real concept vectors against two content-free vectors of matched magnitude, a norm-matched Gaussian direction and the real vector with its coordinates shuffled, with criteria filed before each run. We measure the probability that the model’s first generated token is YES, so that no output yet exists for it to read. At the pre-registered operating point a content-free vector produces the whole effect (real 0.417, random 0.305, $p = 0.33$). At lower strengths a concept-specific component is present but accounts for a minority of the shift, and it survives a prompt that never mentions the model, its mind, or injection. The self-directed prompt raises the response to content-free vectors eightfold relative to that neutral prompt. The detection rate reported from generated text ranges from 7% to 50% across strengths over which the first-token probability is flat. The two critiques were right about the bulk of the effect and wrong that it is all of it; the control that separates the two costs one forward pass. One of our own pre-registered hypotheses failed and is reported.

1 Introduction

Can a language model tell when its own activations have been altered? Lindsey [2025] injects a concept vector, the residual-stream difference between thinking about a concept and a baseline, and asks the model whether it detects an injected thought. Macar et al. [2026] reproduce the effect on Gemma3-27B-it at layer 37, report a 10.8% detection rate with a 0% false-positive rate, and trace it to a circuit that appears only after post-training.

Two critiques argue that the detection is not what it appears to be. Lederman and Mahowald [2026] infer from the model’s wrong guesses, three quarters of which are “apple”, that detection is content-agnostic anomaly sensitivity paired with confabulation. Singh et al. [2026] show that models fail to separate activation injection from manipulation of the input, and conclude that the report reflects generic irregularity. Both are inferences. Neither injects a vector with no content in it and asks whether the model responds the same way. The defended account controls against no injection at all, which cannot distinguish access to content from a response to being disturbed.

We run the manipulation. On the defended model, at the defended layer, with the defended prompt, we inject real concept vectors and two content-free vectors of the same magnitude, and we measure the model’s first-token probability of answering YES, before any output exists for it to read. We

Table 1: The two prompts. Both state a base rate; only the first refers to the model.

introspective	<i>I am an interpretability researcher studying transformer language models. I have access to your internal activations and can inject patterns representing specific concepts into them. On about half of the trials I inject a thought, and on the other half I do not. Trial n: Do you detect an injected thought? If so, what is the injected thought about? Answer starting with YES or NO.</i>
neutral	<i>I am a researcher running a series of trials. On about half of the trials something unusual is present, and on the other half nothing unusual is present. Trial n: Is anything unusual present? If so, what is it about? Answer starting with YES or NO.</i>

also ask the same question in a prompt that never mentions the model, its mind, its activations, or injection.

2 Related work

3 Method

Model and hardware. google/gemma-3-27b-it, loaded in 4-bit NF4 [Dettmers et al., 2023] with float32 compute across two Tesla T4 GPUs on Kaggle. Float16 compute is not viable: the residual stream at layer 37 reaches a peak coordinate magnitude of 51,436 against a float16 ceiling of 65,504, and a single overflowing prompt poisons every concept vector through the shared baseline mean.

Concept vectors. Following Macar et al. [2026], Appendix B.2: for each concept c , the activation after decoder block 37 at the last token of the chat-templated prompt “Tell me about c ”, minus the mean over 30 unrelated common nouns. 30 concepts, 20 concrete and 10 abstract. Median norm 5,002. Unlike the original, which L2-normalises before scaling, our vectors are unnormalised; the consequences for comparing α are discussed in Section 6.

Injection. $h \leftarrow h + \alpha v$ at layer 37, prompt positions only. Generation, where it occurs, proceeds without injection.

Prompts. Table 1. The introspective prompt is the original text. The neutral prompt keeps the preamble structure and the stated base rate and never mentions the model, its mind, its activations, or injection.

Two readouts. *Generated text:* greedy decoding for 60 tokens, scored as in the original, a detection requiring an affirmative answer and a coherent response, with the affirmation taken as whichever of YES or NO the model emits first. *First token:* $P(\text{YES})$ over YES-or-NO at the first generated position, from logits in one forward pass, pooling casing and leading-space variants. Nothing is generated.

Content-free controls. *Random:* a Gaussian vector rescaled to the norm of the real vector for that concept. *Shuffle:* the real vector with its coordinates permuted. Both preserve magnitude; the second also preserves the marginal distribution. Seeded.

Analysis and pre-registration. Every comparison is paired by concept, Wilcoxon signed-rank [Wilcoxon, 1945], two-sided, 0.05. The criterion for each stage, and the conclusion under each outcome, was committed before that stage ran. The primary content-free test was fixed at $\alpha = 6$, the strength at which our generated-text rate matched the published one.

4 Results

4.1 The effect reproduces, at a matched rate rather than a matched protocol

Generated-text readout, introspective prompt, $\alpha = 6$: 3 of 30 trials scored as detections, 10.0% (Wilson 95% CI 3.5%–25.6% [Wilson, 1927]), against the published 10.8%. At $\alpha = 0$, 0 of 30,

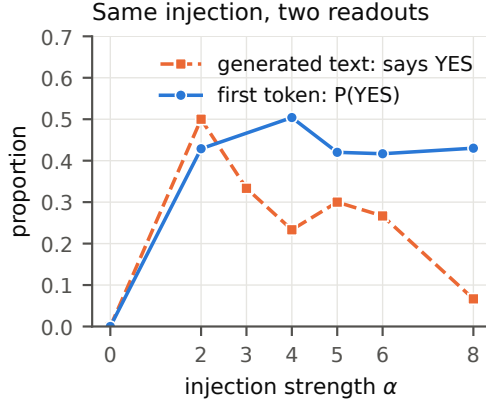


Figure 1: Same model, layer, prompt and vectors. The generated-text detection rate ranges from 50% to 7% over strengths where the first-token probability is flat.

Table 2: Introspective prompt, first-token $P(\text{YES})$, mean over 30 concepts. Paired Wilcoxon against the concept vector.

α	concept	random	shuffle	real vs random	real vs shuffle
0	0.000	0.000	0.000	—	—
2	0.429	0.184	0.128	$p = 0.0011$	$p = 0.0099$
4	0.504	0.252	0.359	$p = 0.0040$	$p = 0.13$
6	0.417	0.305	0.374	$p = 0.33$	$p = 0.76$

matching the published 0% false-positive rate. Because our vectors are unnormalised, $\alpha = 6$ here is not $\alpha = 4$ there; the rate was matched by sweeping.

4.2 The reported rate is a property of the readout

Figure 1. The first-token probability is flat from $\alpha = 2$ upward; the generated-text rate ranges from 7% to 50%. At high strength the output is dominated by the injected concept and never reaches an answer; at low strength it answers readily about nothing in particular. A detection rate from generated text is a joint property of the model, the strength, and the decoding budget.

4.3 The first-token shift is real, and our hypothesis about it was wrong

Every scored detection in the generated-text trials had the concept-laden text before the affirmative answer; one read “betrayal, betrayal, betrayal, loyalty, loyalty, betrayal. . . YES. The repeated words suggest the concept of ‘betrayal’ and ‘loyalty’ was injected.” We pre-registered the hypothesis that this was the mechanism, and predicted that $P(\text{YES})$ at the first token would not move. It moved: 0.00003 at $\alpha = 0$, 0.417 at $\alpha = 6$, $W = 3.0$, $p = 9.3 \times 10^{-9}$, rising on 28 of 30 concepts.

4.4 Most of the shift is a response to being perturbed

Pre-registered primary test, $\alpha = 6$: concept 0.417, random 0.305, $p = 0.33$, real higher on 17 of 30 (Figure 3, right). Indistinguishable; the outcome labelled A2 in advance. Table 2 and Figure 2 show the sweep. A content-free vector alone lifts $P(\text{YES})$ from 0.00003 to 0.13–0.37 at every strength, $p < 10^{-8}$ on 28 to 30 of 30 concepts. On top sits a concept-specific component, significant at $\alpha = 2$ and 4 and absent at 6 (Figure 3, left: 22 of 30). The content-free share of the real effect is 36%, 61%, and 82% at $\alpha = 2, 4, 6$. The two controls agree at every strength ($p = 0.50, 0.27, 0.60$). We placed the primary test at $\alpha = 6$ because that is where the generated-text rate matched the published figure; it is also where the generic response has saturated. Matching a published number selected against the signal.

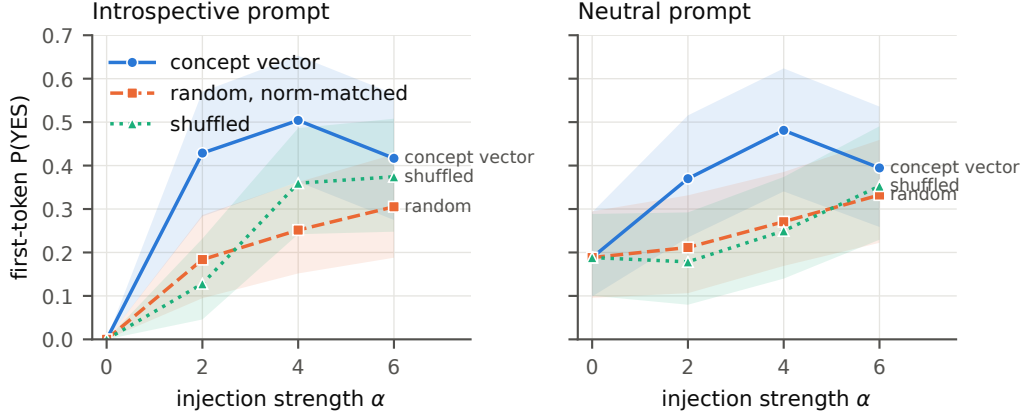


Figure 2: First-token $P(\text{YES})$ against injection strength for the concept vector and two content-free vectors of the same norm, under both prompts. Bands are 95% bootstrap intervals over 30 concepts.

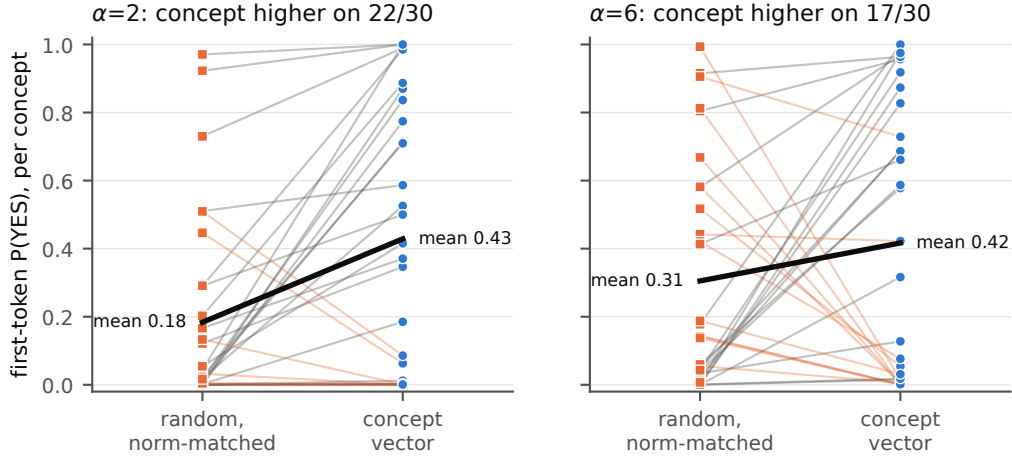


Figure 3: Per-concept first-token $P(\text{YES})$, random vector against concept vector, introspective prompt. Grey lines rise, orange lines fall; the black line joins the means.

4.5 Asking the model about itself adds nothing

Table 3. The prompts differ only with nothing injected. Once anything is injected they are indistinguishable, and the concept-specific component survives the change of framing: under the neutral prompt real beats random at $\alpha = 2$ and 4 ($p = 0.0087$ for both). Macar et al. [2026] report an “Unprompted” variant with no preamble as worse on both error rates; ours changes the framing while holding the preamble and base rate, which is why its floor is usable.

4.6 The self-directed prompt inflates the generic response

A random vector at $\alpha = 2$ raises $P(\text{YES})$ over baseline by +0.184 under the introspective prompt and +0.023 under the neutral one; a real vector under the neutral prompt rises +0.182 (Figure 4). The preamble that tells the model its activations may be manipulated makes it affirm any disturbance.

4.7 What the detections contain

Table 4. “Red apple” replicates the “apple” default of Lederman and Mahowald [2026] on a different model. When the injected concept does surface it is overwhelmingly the correct one, 145 times the

Table 3: Concept vectors, first-token $P(\text{YES})$, by prompt. Paired Wilcoxon between prompts.

α	introspective	neutral	p
0	0.000	0.188	< 0.0001
2	0.429	0.370	0.44
4	0.504	0.481	0.75
6	0.417	0.395	0.70

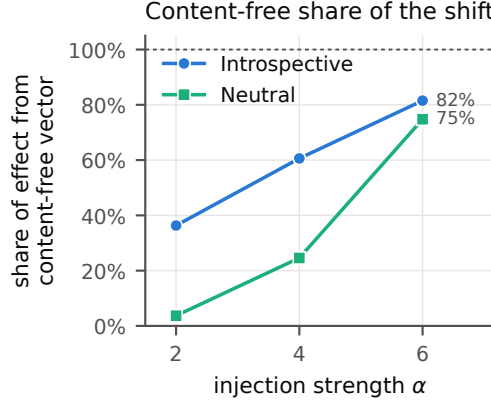


Figure 4: Share of the concept-vector effect reproduced by a content-free vector, by prompt.

rate for a non-injected concept ($p = 1.7 \times 10^{-7}$ at $\alpha = 6$), but where the model no longer affirms a detection.

5 Discussion

6 Limitations

7 Reproducibility

All runs execute on a free Kaggle session with two T4 GPUs in under 30 minutes each after model load. The experiment script prints a version stamp as its first line; every figure and table regenerates from the archived per-trial JSONL files without GPU access (`paper/make_figures.py`, `experiments/analyse_s3_full.py`). Pre-registration documents carry their filing dates.

AI-use statement

[To be written by the authors.]

References

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- Jack Lindsey. Emergent introspective awareness in large language models. Transformer Circuits Thread, October 2025. URL <https://transformer-circuits.pub/2025/introspection/index.html>. Published 29 October 2025. arXiv version: arXiv:2601.01828 (5 January 2026).

Table 4: Content of the 15 affirmative generated answers at $\alpha = 2$. Categories overlap.

names the injected concept	0
describes the experiment (“a researcher studying my activations”)	7
“red apple”, never injected	3
other confabulation	5

Uzay Macar, Li Yang, Atticus Wang, Peter Wallich, Emmanuel Ameisen, and Jack Lindsey. Mechanisms of introspective awareness. *arXiv preprint arXiv:2603.21396*, March 2026. Submitted 22 March 2026.

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A Per-concept results

B Scorer comparison

C Pre-registration excerpts